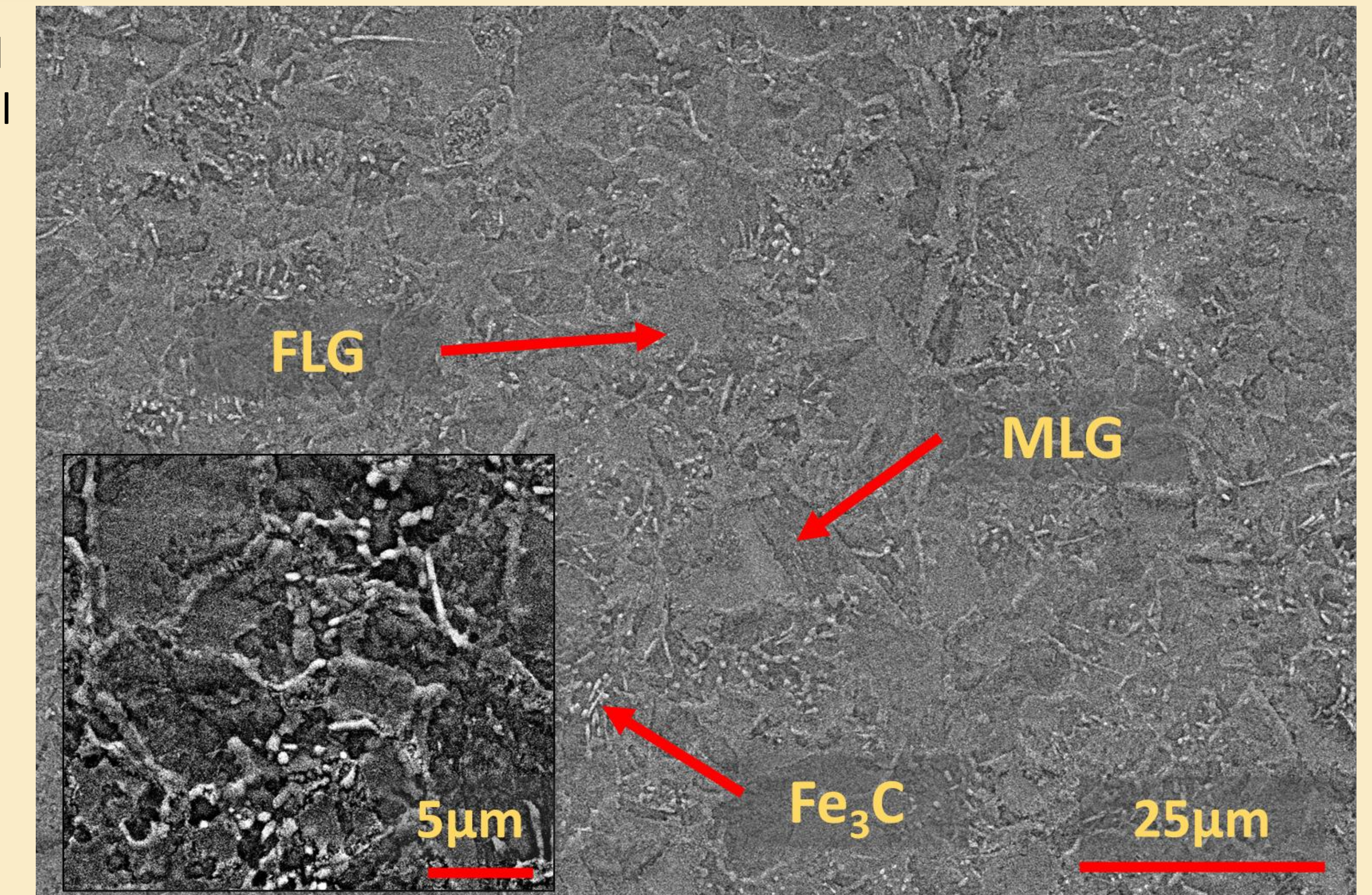


Introduction

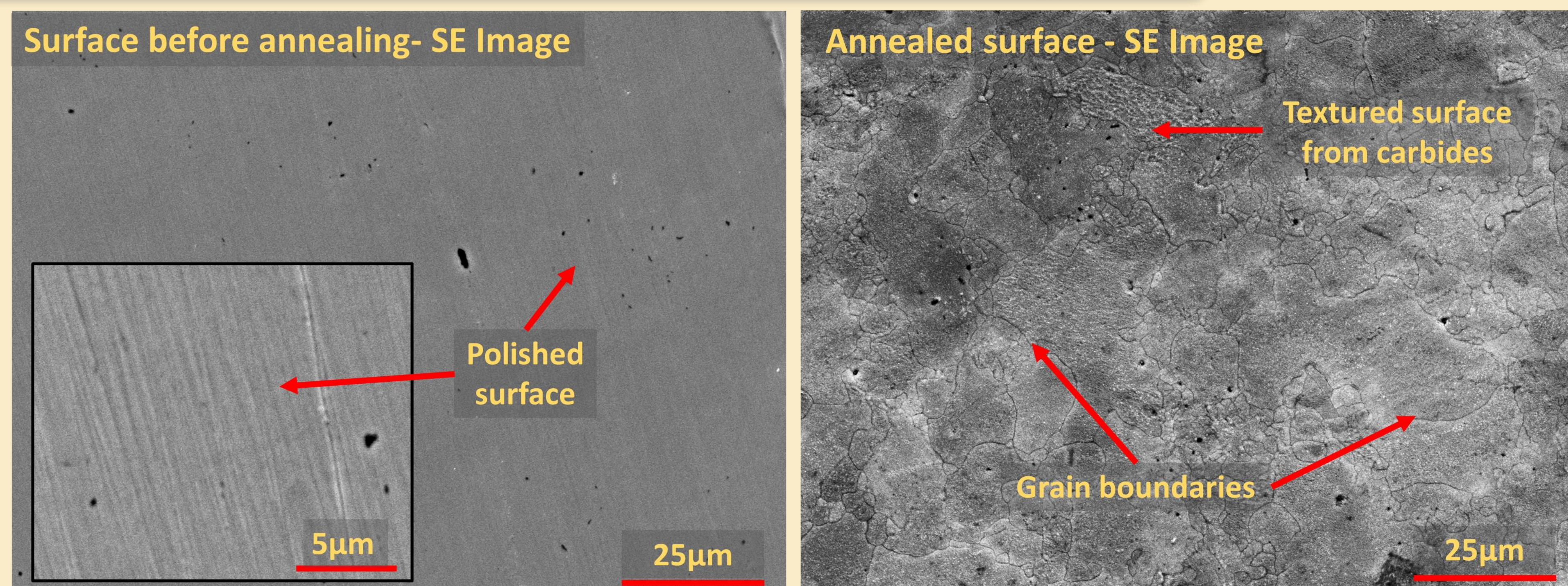
- This study focuses on surface microstructures for 8620 steel before and after different annealing conditions to improve graphene surface coatings, toward the goal of creating hydrophobic, corrosion resistant graphene coatings.
- Samples were vacuum annealed at 650°C and <50mTorr for 60 minutes under two conditions, H₂ and Ar or H₂ and C₂H₂.
- After annealing in an H₂ and Ar environment, the surface polishing artifacts smoothed, grain size increased, and carbides precipitated on some surfaces.
- In the H₂ and C₂H₂ annealing environment, graphene formed on the surface. The hydrogen maintained a reducing atmosphere and increases the reactivity of the steel surface, and the C₂H₂ increased the carbon content of the surface.
- With H₂ and C₂H₂ annealing, the surface microstructure changed to support graphene growth, and carbon precipitated on the steel surface as multilayer graphene (MLG), few layer graphene (FLG) and single layer graphene (SLG).

Graphene coatings after annealing with H₂ and C₂H₂

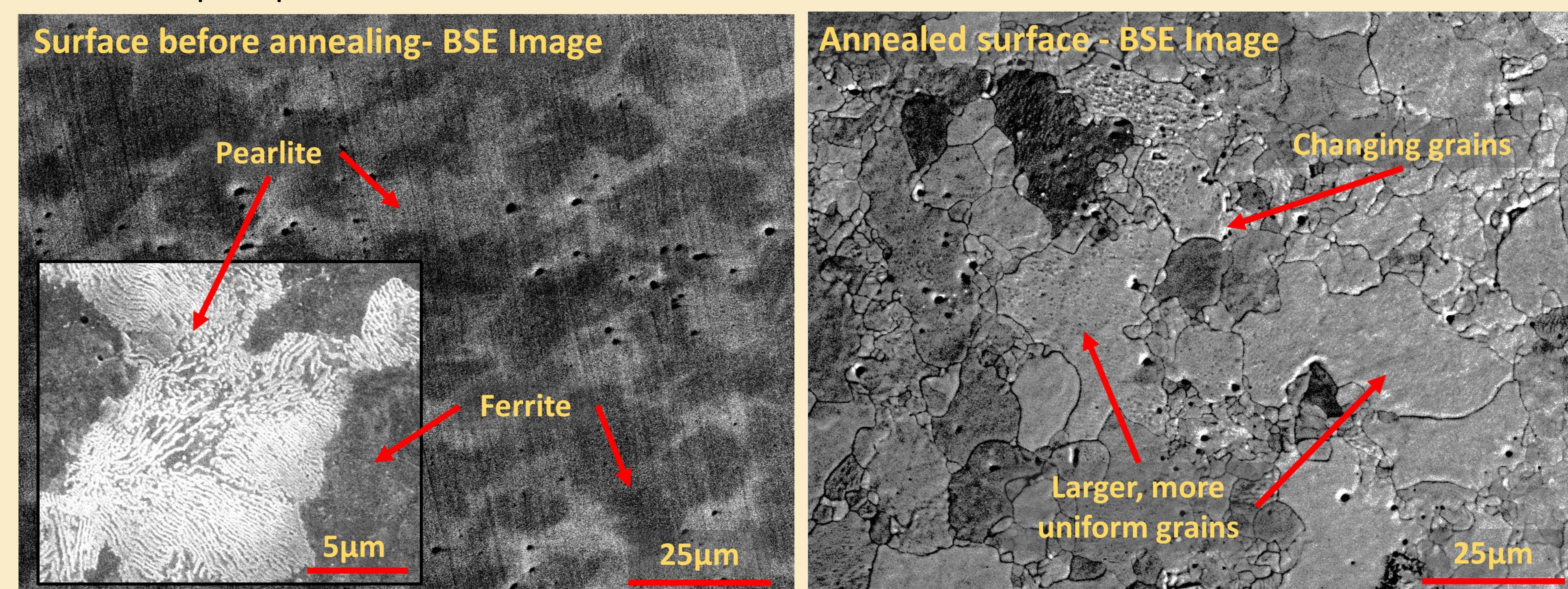
- The high magnification SEM image on the right shows detail of the graphene coating formed on the surface of 8620 steel during annealing with acetylene.
- Annealing 8620 with acetylene increases surface carbon content, resulting in a primarily MLG (dark areas) coating, with FLG covering some areas (light areas).
- The high surface carbon content results in Fe₃C forming on some of the surface (white areas), especially at grain boundaries.
- The graphene growth is influenced by the surface microstructures of the steel, which are altered by annealing with acetylene to support graphene growth.



Surface Microstructures after Annealing under H₂ and Ar

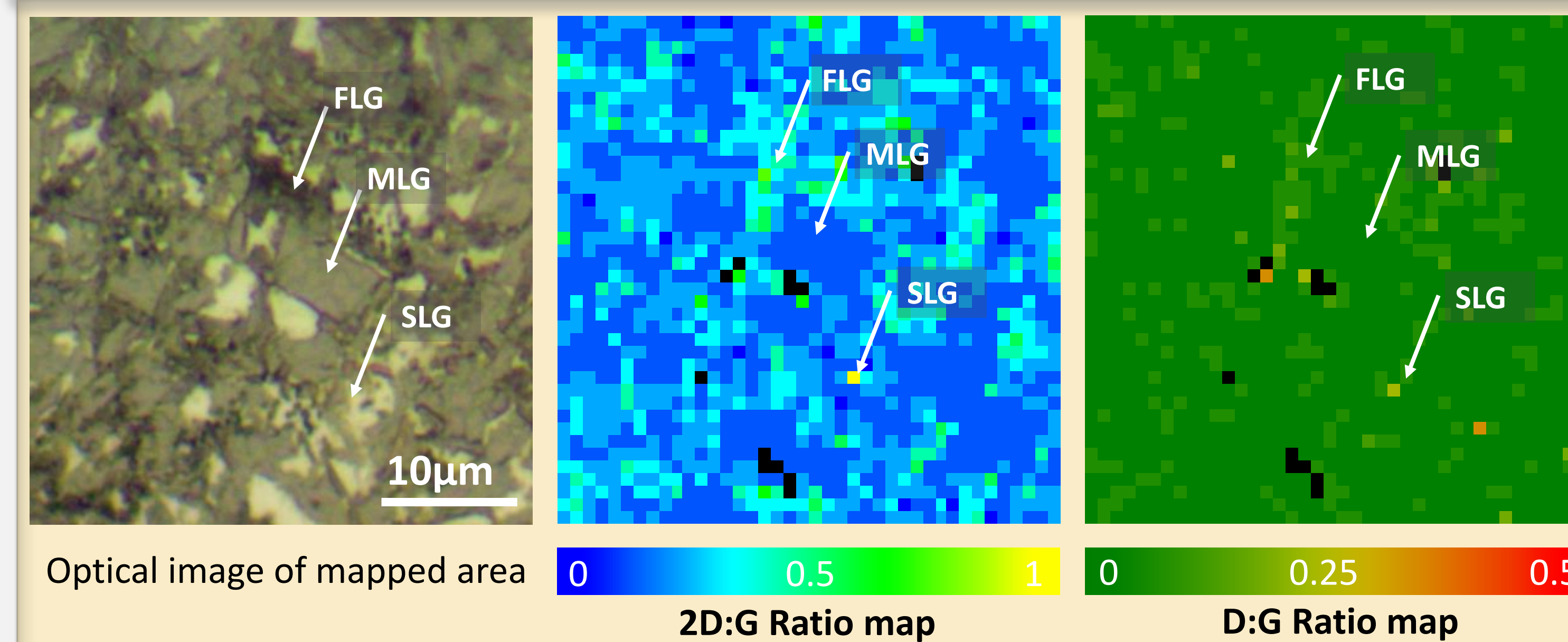


- The SE image on the left shows the polished 8620 steel surface before annealing, which is flat with some polishing scratches and inclusions visible. On the annealed surface (right), scratches were gone, grain boundaries were visible, and the surface morphology changed where carbides precipitated.

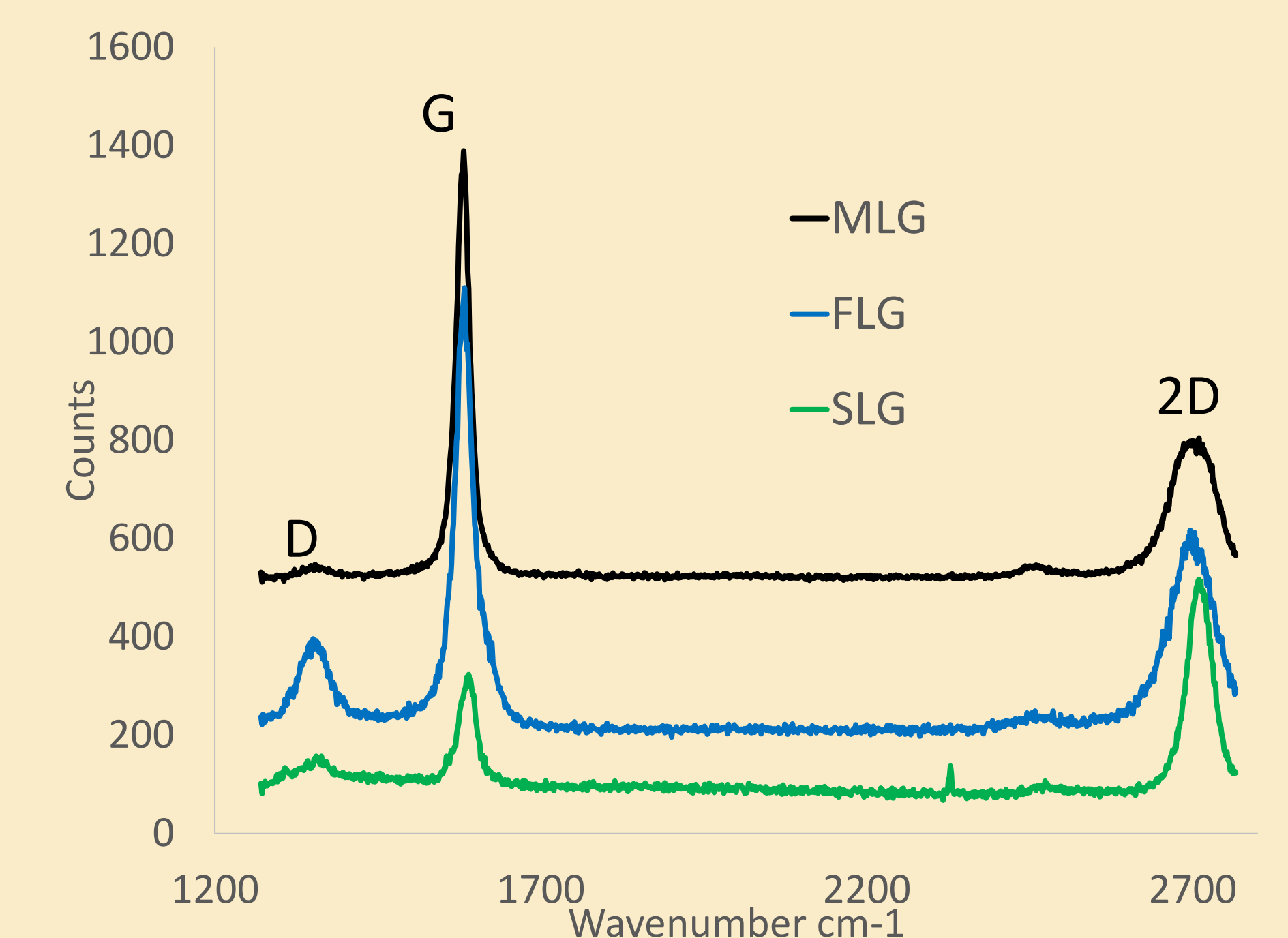


- The BSE image on the left reveals the ferrite and pearlite beneath the surface of the polished sample before annealing (grains 5-15µm in size). The inserted SE image shows typical pearlite in this steel. In the annealed sample (right), grains were larger (10-30µm in size), boundaries were more visible, and the contrast shows a more uniform annealed microstructure.

Mapping graphene coating coverage with Raman spectroscopy

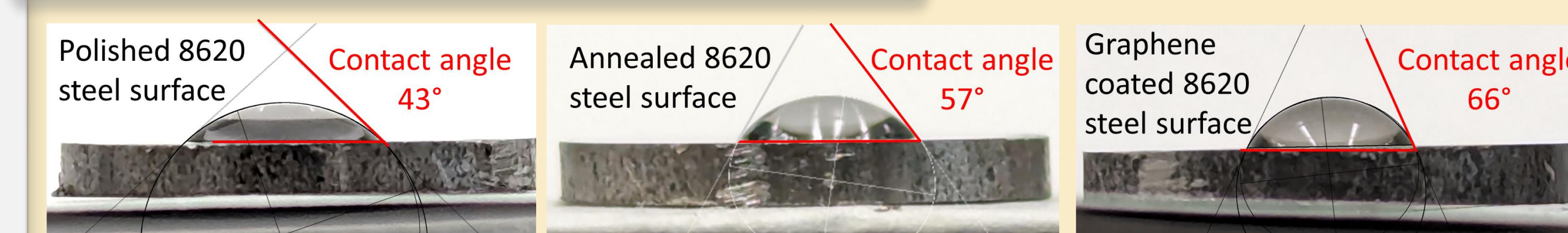


- Raman maps of the 30µm x 30µm area in the optical image (900 spectra) show the coating quality.
- In the 2D:G ratio map blue pixels show MLG, and teal shows FLG. Some SLG is present in the coating gaps. Black pixels show an uncoated area.
- The D:G ratio map shows that most of the coating had a very low D:G ratio of ~0.1 or lower.
- The coatings had coverage >95% and uniformly covered samples 1cm in diameter.



- The typical Raman spectra for the graphene coatings are shown below
- The 2D to G peak ratio is a measure of graphene thickness, which is >1 for SLG, and <1 for FLG and MLG.
- The D to G peak ratio shows disorder in the graphene coating. The low ratio of <0.1 for the MLG shows very low disorder, and the FLG and SLG show a higher D:G peak ratio of 0.2 and 0.46 respectively.

Contact Angle Measurement



- The larger the contact angle of a water droplet, the more hydrophobic a steel surface is.
- The graphene coated surface was the most hydrophobic with the largest contact angle. This indicates the graphene coated surface will be the most corrosion resistant.

Conclusion

- This study demonstrates that annealing 8620 steel at 650°C transforms the surface microstructure.
- Annealing under H₂ and Ar smooths surface artifacts, exposes grain boundaries, and results in larger, more uniform grains with surface carbides.
- Annealing under H₂ and C₂H₂ increases surface carbon content and results in a primarily MLG and FLG coating, with SLG coating some gaps.
- The graphene coated surfaces were the most hydrophobic, forming corrosion resistant coatings.